

BNY MELLON – INTERNSHIP INTERVIEW EXPERIENCE

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The interview process consisted of 3 rounds:

- 1. Online Assessment**
- 2. Technical Interview**
- 3. Technical & HR Interview**

First Round – Online Assessment

The first round was an online coding assessment. It consisted of 4 coding questions **2 medium** and **2 hard** and the total time given to solve them was **2 hours and 45 minutes**.

- **Question 1: Optimal Team Selection (Medium)**

You are given n developers with unique skill levels and two arrays: `lowerSkill` and `higherSkill`.

`lowerSkill[i]` shows how many teammates with lower skill a developer can work with.

`higherSkill[i]` shows how many teammates with higher skill a developer can work with.

You need to find the largest possible team where every developer's limits are satisfied.

Sample Input: $n = 6$, `lowerSkill = 1,1,2,2,3,5`, `higherSkill = 5,4,3,2,1,0`

Sample Output: 5

Explanation: The largest team that satisfies everyone's limits has 5 developers.

- **Question 2: Maximum Redundancy of Servers (Medium)**

You are given a binary string where 1 means the server is working and 0 means it is not. You can choose one continuous segment of servers and flip them (0 to 1 and 1 to 0).

You need to find the maximum number of different possible counts of working servers after flipping at most once.

Sample Input: servers = 010

Sample Output: 3

Explanation: After flipping the best segment, there are 3 possible counts of working servers.

Sample Input: servers = 111

Sample Output: 4

Explanation: Flipping one segment can give 4 different possible counts of working servers.

- **Question 3: Largest High-Priority Subset of Events (Hard)**

You are given n events with start and finish times. A high-priority subset is a group of events where at least one event overlaps with every other event in the group. Find the largest size of such a subset.

Sample Input: $n = 4$, start = 1,2,3,4, finish = 2,3,5,5

Sample Output: 3

Explanation: The largest subset where one event overlaps with all others has 3 events.

- **Question 4: (Hard)**

I don't remember the exact problem statement.

Performance in the Round:

Question 1 – 15/15 test cases passed

Question 2 – 10/15 test cases passed

Question 3 – 10/15 test cases passed

Question 4 – 9/15 test cases passed

After this, a total of 26 students were shortlisted for the next round.

Second Round – Technical Interview

The second round was a **technical interview conducted in CUIC**. A total of 26 students were **split across 6 panels**, and I was called **second** in my panel. Each panel had **one interviewer**. My panel was very friendly and chill, which helped me feel comfortable initially.

After settling in, the interview started with a **resume discussion**:

- He saw my name and said that he has a friend with the same name
- He asked where I was from.
- Seeing that I mentioned I know Kannada, he asked how I knew it and whether I was fluent.
- We had some **small talk** before moving to technical questions.

Next, I was asked to:

- Introduce myself.
- Explain one project mentioned in my resume.
- Draw the schema and relationships among tables in another project.
- Explain how I normalized my tables.
- Write SQL queries on the given tables.
- Answer why I used MySQL in one project and MongoDB in another.
- List SQL inbuilt functions I knew, such as aggregate, date, and string functions.

Then, the interview moved to DSA concepts:

- I was asked about data structures like stack and queue, including functions like push, pop, empty.
- After explaining, the interviewer asked me to assume the listener doesn't know coding and explain the concept again. I took some time and explained it clearly.

After that, I was asked to choose a programming language I'm comfortable with. I chose **C++**, and they gave me an easy coding problem:

- Three people are investing in one product: A = 1200, B = 800, C = 1000.
- Total = 3000, so each person's share = 1000.
- Calculate how much extra or less each contributed and return an array like [200, -200, 0].
- While writing the code, I explained my logic. Even if it was wrong, I did this because **interviewers want to understand your thinking process**, so always speak your logic while coding.

Finally:

- I was asked if I had any questions for them, and I asked them "Do interns need any prior knowledge to perform well in the role?" and "What challenges would an intern face during the internships?"
- The interview went very well.

Out of 26 students, **14 were selected for the next round**, and I was **one of them**.

Third Round – Technical + HR Interview

The third round was conducted on CUIC. This time, each student had a different panel. I was interviewed individually.

The interview started with self-introduction.

After that:

- The interviewer asked about my project, and I explained it.
- She asked about my role, and I said I worked on frontend, database, and understood backend logic.
- She asked how API calls work?
- I explained that we used OpenAI for summarization.
- She asked how I implemented it and what model I used, I said GPT-4o mini, and she asked why that model. she asked me to write backend code
- She asked what I would do if an API call failed. I said we could display a user-friendly message.
- She asked how to improve that functionality, and I suggested using a backup API if the first one fails.
- She asked what if the backup API also fails.
- She asked me to **write all code on paper**. I explained that since I worked in frontend, I didn't know exact backend syntax, but I tried to implement the logic, and if both APIs fail, I said I would use a frontend library for summarization, but she insisted on backend implementation, which I managed.
- JWT – why it was used and write code to generate a JWT token.
- Role-based rendering – how pages render for student, faculty, admin, writing the code for role-based access.
- Handling wrong registration – e.g., if a student tries to register as faculty. I explained the verification process by admin.
- Why the functionality was developed – to implement course enrolment logic since our department didn't have it.

She also gave a coding problem:

- Write a program from scratch to find the second largest element in an array. I implemented it from input to output.

She asked some HR questions like:

- Challenges faced while working on the project.

Finally, the interviewer asked, “Any questions for me?” and I asked them “What challenges did you face when working for BNY?” and “What feedback would you give me and if I can improve in any areas?”

After this round, **6 students were selected for the internship** from CEG and MIT.

Key Takeaways:

- Problem Solving & DSA: Start solving problems as early as you can. In interviews, start with a brute-force approach first and then gradually optimize to an efficient solution.
- Have a clear understanding and hands-on experience with all the projects you have worked on. This plays a crucial role in interviews.
- Be ready to answer all kinds of questions from your resume confidently.
- Be strong in DBMS, OOPs, and other core subjects, as technical questions can cover these areas.
- If you don't know the exact answer, don't just say "I don't know." Try to reason it out or provide a logical approach, even if it's not fully correct, interviewers want to see your thinking process.

Don't get discouraged if you don't get shortlisted. Keep putting your best efforts, stay consistent, and never lose hope. The right opportunity will come to you, just keep trying and trust your journey!

All the very best!!!

Feel free to reach out me:

Email: s4tech1234@gmail.com

Phone: +91 9962767447

Best Regards,

Saravana Kumar B